Make Spatial Cross-Validation Assignments

Description

Assigns complete sampling locations to deterministic spatially stratified folds while balancing grid-cell coverage, location counts, and sample counts.

Usage

```
build_spatial_cross_validation_assignments(
  data_locations = NULL,
  n_folds = 5L,
  n_repeats = 1L,
  grid_cell_size_km = NULL,
  seed = 900723L
)
```

Arguments

`data_locations`	Location table returned by [build_cross_validation_location_table()] with projected coordinates, sample counts, and sample-row indices.
`n_folds`	Single integer greater than one and no greater than the location count.
`n_repeats`	Single positive integer. Grid origins and within-cell ordering change deterministically between repeats. Defaults to '1L'.
`grid_cell_size_km`	Single positive numeric grid-cell width in projected kilometres.
`seed`	Single finite integer used for reproducible within-cell ordering. Defaults to '900723L'.

Details

Locations in the most populated cells are assigned first. For each location, folds are ranked lexicographically by existing locations from the same grid cell, total locations, total represented samples, and fold ID. Grid origins shift deterministically between repeats to expose sensitivity to arbitrary grid boundaries. The caller's random-number state is restored.

Value

Tibble with 'repeat_id', 'fold_id', 'location_id', 'grid_cell_id', 'n_samples', and 'row_indices'. Every location occurs exactly once per repeat and every sample row remains attached to its complete location.

Examples

```
data_locations <-
  tibble::tibble(
    location_id = c("a", "b", "c", "d"),
    coord_x_km = c(0, 1, 10, 11),
    coord_y_km = c(0, 1, 10, 11),
    n_samples = rep(1L, 4L),
    row_indices = list(1L, 2L, 3L, 4L)
  )
build_spatial_cross_validation_assignments(
  data_locations = data_locations,
  n_folds = 2L,
  grid_cell_size_km = 5
)
```